FORCES AND MOTION

GUIDED NOTES · PHASE 02 · 6 SECTIONS · TEACHER KEY

NAME _____ DATE _____ PERIOD _____

UNIT LEARNING TARGETS

- I can state Newton's three laws and use Fg = mg to find an object's weight.
- I can draw a free-body diagram and find the net force on an object, including forces at an angle.
- I can split an object's weight into parts along and into a ramp and find the normal force.
- I can use Fnet = ma to find acceleration, including on a ramp and in an elevator.
- I can tell static from kinetic friction and find friction, net force and acceleration on level ground and on a ramp.
- I can find tension and acceleration for objects joined by cables, ropes and pulleys.

KEY TERMS

- **S03.1** Force, newton (N), mass, weight, equilibrium, net force
- **S03.2** Free-body diagram, normal force, tension, resultant, components
- **S03.3** Incline, Fg,parallel, Fg,perp
- **S03.4** Acceleration, apparent weight
- **S03.5** Static friction, kinetic friction, coefficient of friction (μ)
- **S03.6** Ideal string, pulley, Atwood machine

HOW THESE NOTES WORK

- The left column names what to record and the slide it comes from.
- Fill each blank while that slide is up.
- A gray rule down the left of a line marks a must-write line.
- In each work box, state your positive direction and carry units to the answer.

SECTIONS IN THIS UNIT

		DIFFICULTY
☐	S03.1 Newton's Laws	**3 / 10**
☐	S03.2 Free-Body Diagrams	**6 / 10**
☐	S03.3 Forces on an Incline	**5 / 10**
☐	S03.4 Net Force and Acceleration	**6 / 10**
☐	S03.5 Friction	**7 / 10**
☐	S03.6 Connected Objects	**8 / 10**

Estimated difficulty: 1 = introductory; 10 = highly challenging.

EQUATION TOOLBOX

Newton's laws

$F_g = mg$

$F_{net} = ma$

$a = \frac{F_{net}}{m}$

$F_N - mg = ma$ (up is positive)

Components

$F_x = F\cos\theta$

$F_y = F\sin\theta$

$F_{net} = \sqrt{(F_x^2 + F_y^2)}$

$\theta = \tan^{-1}(F_y/F_x)$

Ramps and friction

$F_{g,parallel} = mg\sin\theta$

$F_{g,perp} = F_N = mg\cos\theta$

$f_{s,max} = \mu_s F_N$

$f_k \approx \mu_k F_N$

$g = 9.8\ m/s^2$ on Earth. $1\ N = 1\ kg\cdot m/s^2$. μ has no units. Frictionless ramp: $a = g\sin\theta$.

Newton's Laws

LEARNING TARGET I can state Newton's three laws and use Fg = mg to find an object's weight.

FORCE · MASS AND WEIGHT

Slides 3–4

Common mix-up: mass (kg) and weight (N) are different quantities.

A force is a **push** or a **pull**: a **vector**, with size and **direction**, in **newtons (N)**.

Mass: how much **matter**, in **kilograms (kg)**. Stays the same when you travel. **Weight:** the force of **gravity**, in **newtons (N)**. Changes when gravity changes.

10 kg on Earth: Fg = 10(9.8) = **98 N** down. 10 kg on the Moon: Fg = 10(1.6) = **16 N** down.

> **Weight: Fg = mg. On Earth, g = 9.8 m/s².**

WEIGHT ON EARTH AND THE MOON

Slide 5

Same mass, different gravity

Given: m = 60 kg; Earth: g = 9.8 m/s²; Moon: g = 1.6 m/s². **Find:** weight in each location.

WORK / show your reasoning and units

Fg = mg

Earth: Fg = 60(9.8) = 588 N down

Moon: Fg = 60(1.6) = 96 N down

Same mass. Different gravity. Different weight.

Earth: Fg = 588 N down **Moon: Fg = 96 N down**

NEWTON'S THREE LAWS

Slides 6–8

Common mix-up: a moving object does not need a net force to keep moving. A moving puck keeps going with Fnet = 0.

> **1st law: an object at rest stays at rest, and an object in motion keeps a constant velocity, unless a net external force acts on it.**

Equilibrium: forces balance, Fnet = **0**. At rest, or constant speed in a **straight line**.

> **2nd law: Fnet = ma, so a = Fnet / m. Fnet is the sum of all forces.**

Same mass: more net force, **more** acceleration. Same force, twice the mass: **half** the acceleration.

> **3rd law: for every action, there is an equal and opposite reaction.**

The pair acts on **two** objects, so third-law forces do not **cancel** on one object's FBD.

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

State Newton's three laws in your own words, and write the equation for weight.

1st: no net force, no change in velocity. 2nd: Fnet = ma. 3rd: forces come in equal, opposite pairs on two objects. Weight: Fg = mg.

- [ ] I can tell mass (kg) from weight (N) and find weight with Fg = mg.
- [ ] I can state all three laws and explain equilibrium.

Free-Body Diagrams

LEARNING TARGET I can draw a free-body diagram and find the net force on an object, including forces at an angle.

FREE-BODY DIAGRAMS · NORMAL FORCE

Slides 10–11

Common mix-up: FN does not always equal weight. That needs the level-surface conditions above.

Draw the object as a **box** or **dot**. Add one labeled **arrow** for each force. Longer arrows mean **larger** forces. Keep **motion** arrows separate.

Fg: gravity pulls **down**. FN: surface pushes **perpendicular**. T: rope pulls **along the rope**.

> **Show only the forces that act on the chosen object.**

Normal means **perpendicular**. The normal force is a surface's **push** on an object. It points straight out from the surface: on a floor, **up**; on a ramp, **tilted**.

> **On a level surface with no other vertical forces or acceleration: FN = Fg.**

Extra notes:

BALANCED FORCES: BOOK AND LAMP

Slide 12

Two FBDs: book and lamp

Given: 2 kg book at rest on a table; 1.5 kg lamp at rest on a rope; $g = 9.8\ m/s^2$.

Find: the free-body diagram of each, and the upward force on each.

WORK / show your reasoning and units

Book FBD: FN up, Fg down, equal arrows. Lamp FBD: T up, Fg down, equal arrows.

Book: FN = Fg = 2(9.8) = 19.6 N up

Lamp: T = Fg = 1.5(9.8) = 14.7 N up

Book: FN = 19.6 N up Lamp: T = 14.7 N up

NET FORCE IN ONE DIMENSION

Slide 13

Choose a positive direction, then add forces with **signs**. Add horizontal and vertical forces **separately**.

Given: crate; 30 N right; 10 N left. Positive: right. **Find:** Fnet, with its direction.

WORK / show your reasoning and units

Right: +30 N Left: –10 N

Fnet = 30 – 10 = 20 N

20 N to the right

Fnet = 20 N to the right

RESULTANT FORCE AND DIRECTION

Slide 15

Given: 40 N right (+x) and 30 N up, perpendicular. **Find:** the size and direction of the resultant.

WORK / show your reasoning and units

$F_{net} = \sqrt{(40^2 + 30^2)} = 50$ N

$\theta = \tan^{-1}(30/40) = 36.9°$

50 N, 36.9° above +x

F_{net} = 50 N at θ = 36.9° above +x

BREAKING A FORCE INTO COMPONENTS

Slide 16

Horizontal: F_x = **$F \cos\theta$** Vertical: F_y = **$F \sin\theta$**

Given: F = 100 N pulling 30° above the horizontal. **Find:** the horizontal and vertical components.

WORK / show your reasoning and units

$F_x = F \cos\theta = 100 \cos 30° = 86.6$ N

$F_y = F \sin\theta = 100 \sin 30° = 50.0$ N

Check: $\sqrt{(86.6^2 + 50.0^2)} \approx 100$ N

F_x = 86.6 N horizontal F_y = 50.0 N up

TWO ANGLED FORCES: TUGBOATS

Slide 18

Check the quadrant before reporting an angle

Given: top view; 500 N at 30° above +x; 400 N at 45° above –x. Positive: +x right, +y up the page.

Find: the net force on the barge and its direction.

WORK / show your reasoning and units

$F_x = 500 \cos 30° - 400 \cos 45° = 150$ N

$F_y = 500 \sin 30° + 400 \sin 45° = 533$ N

$F_{net} = \sqrt{(150^2 + 533^2)} \approx 554$ N

$\theta = \tan^{-1}(533/150) \approx 74°$

$F_{net} \approx$ 554 N at $\theta \approx$ 74° above +x

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

Explain how to find the net force on an object from its free-body diagram, including a force at an angle.

Draw one arrow per force on the object. Choose a positive direction. Split angled forces into $F_x = F \cos\theta$ and $F_y = F \sin\theta$, add each direction with signs, then combine with $\sqrt{(F_x^2 + F_y^2)}$ and $\theta = \tan^{-1}(F_y/F_x)$.

- ☐ I can draw an FBD with one labeled arrow per force.
- ☐ I can add forces with signs and split an angled force into components.

Forces on an Incline

LEARNING TARGET I can split an object's weight into parts along and into a ramp and find the normal force.

GRAVITY ON A RAMP · NORMAL FORCE ON AN INCLINE

Slides 20–21

Common mix-up: on a ramp FN is mg cos θ, less than the weight.

Use one axis **along** the ramp and one **perpendicular** to it.

> **$F_{g,parallel} = mg \sin\theta$ down the slope. $F_{g,perp} = mg \cos\theta$ into the surface.**

> **$F_N = mg \cos\theta$ (no extra perpendicular force and no perpendicular acceleration)**

Steeper ramp, **less** support. 15° slope: FN ≈ **0.97mg**. 60° slope: FN = **0.50mg**.

INCLINE EXAMPLE: A 5 KG BLOCK

Slide 22

Given: m = 5 kg; frictionless 30° ramp; g = 9.8 m/s². **Find:** FBD, $F_{g,parallel}$, $F_{g,perp}$, and FN.

WORK / show your reasoning and units

FBD: Fg down, FN perpendicular out of the ramp; Fg split into $F_{g,parallel}$ and $F_{g,perp}$.

Fg = mg = 5(9.8) = 49 N

$F_{g,parallel}$ = 49 sin 30° = 24.5 N down the ramp

$F_{g,perp}$ = 49 cos 30° = 42.4 N into the ramp

FN = 42.4 N, perpendicular out of the ramp

$F_{g,parallel}$ = 24.5 N down the ramp $F_{g,perp}$ = 42.4 N into the ramp FN = 42.4 N out

INCLINE EXAMPLE: A PUSH UP THE RAMP

Slide 23

Given: m = 12 kg; frictionless 25° ramp; 30 N push uphill. Positive: downhill.

Find: $F_{g,parallel}$ and Fnet.

WORK / show your reasoning and units

$F_{g,parallel}$ = 12(9.8) sin 25° = 49.7 N downhill

Fnet = 49.7 − 30 = 19.7 N

19.7 N down the ramp

$F_{g,parallel}$ = 49.7 N downhill Fnet = 19.7 N down the ramp

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

Explain how to split the weight on a ramp, and how FN changes as it gets steeper.

Tilt the axes. $F_{g,parallel} = mg \sin\theta$ down the slope; $F_{g,perp} = mg \cos\theta$ into the ramp; $F_N = mg \cos\theta$. A steeper ramp gives a smaller FN.

- [ ] I can write $F_{g,parallel} = mg \sin\theta$ and $F_{g,perp} = mg \cos\theta$.
- [ ] I can find FN and the net force along a ramp.

Net Force and Acceleration

LEARNING TARGET I can use Fnet = ma to find acceleration, including on a ramp and in an elevator.

CALCULATING ACCELERATION

Slide 25

1 N = **1 kg·m/s²**. Acceleration points in the direction of the **net force**.

Given: m = 4 kg; Fnet = 12 N to the right. **Find:** the acceleration.

WORK / show your reasoning and units

Fnet = ma, so a = Fnet / m

a = 12 N / 4 kg

a = 3.0 m/s² right

a = 3.0 m/s² right

A CAR ACCELERATES

Slide 26

Given: m = 1500 kg; 4500 N forward; 1200 N drag (backward). Positive: forward. **Find:** Fnet and a.

WORK / show your reasoning and units

Fnet = 4500 – 1200 = 3300 N forward

a = Fnet / m = 3300 / 1500

a = 2.2 m/s² forward

Fnet = 3300 N forward a = 2.2 m/s² forward

A SLED PULLED AT AN ANGLE

Slide 28

FN here is less than mg

Given: m = 20 kg; 60 N pull at 35° above horizontal; ignore friction. Positive: +x (right) and up.

Find: the acceleration and the normal force.

WORK / show your reasoning and units

Fx = 60 cos 35° = 49.1 N

a = Fx / m = 49.1 / 20 = 2.46 m/s² right

Vertical: FN + 60 sin 35° – mg = 0

FN = 196 – 34.4 ≈ 162 N up

a = 2.46 m/s² right FN ≈ 162 N up

ACCELERATION DOWN A FRICTIONLESS RAMP

Slide 29

Without friction, a = **g sin θ**. The acceleration does not depend on the block's **mass**.

Given: m = 8 kg; frictionless 20° ramp. Positive: downhill. **Find:** the acceleration.

WORK / show your reasoning and units

Fnet = mg sin θ

ma = mg sin θ

Mass cancels: a = g sin θ = 9.8 sin 20°

a = 3.35 m/s² downhill

a = 3.35 m/s² downhill

ELEVATORS AND APPARENT WEIGHT

Slide 30

Common mix-up: FN equals mg only when a = 0.

A scale reads the upward **normal force**, FN. That is your **apparent weight**.

> **Choose up as positive: FN – mg = ma**

Rest or constant velocity: a = **0**, so FN = **mg**. The scale reads your usual weight.

> **The scale reading changes with acceleration, not simply with motion up or down.**

Extra notes:

ELEVATOR ACCELERATING UP

Slide 31

Given: m = 70 kg; a = 2.0 m/s² upward. Positive: up. **Find:** the scale reading.

WORK / show your reasoning and units

Weight: mg = 70(9.8) = 686 N

FN – mg = ma

FN = 686 + 70(2.0)

FN = 826 N (up)

FN = 826 N

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

Explain the steps from a free-body diagram to an acceleration, and why a scale reads differently in an accelerating elevator.

Choose a positive direction, add the forces to get Fnet, then a = Fnet / m in the direction of Fnet. The scale reads FN, and FN – mg = ma, so FN changes when a ≠ 0.

- [ ] I can find a from Fnet = ma, with units and direction.
- [ ] I can find a scale reading with FN – mg = ma.

Friction

LEARNING TARGET I can tell static from kinetic friction and find friction, net force and acceleration on level ground and on a ramp.

STATIC AND KINETIC FRICTION · COMPARING COEFFICIENTS

Slides 34–35

Table values are approximate (OpenStax)

Friction opposes **sliding**, or the tendency to slide, between surfaces.

> **Static (not sliding): fs adjusts to match the needed force. Maximum: fs,max = μsFN**

> **Kinetic (sliding): fk ≈ μkFN. Usually smaller than maximum static.**

At rest with a 20 N push: fs = **20 N**, opposite the push. Keep sliding: use **μk**. μ has no **units**.

Same normal force: higher μk means **more** sliding friction. Shoes on ice: μk = **0.05**. Synovial joints: μk = **0.015**. Your knee slides with less friction than shoes on ice.

Extra notes:

FRICTION AT CONSTANT VELOCITY

Slide 36

Given: m = 40 kg; level floor; constant velocity; 120 N horizontal push. **Find:** FN, fk, and μk.

WORK / show your reasoning and units

Vertical balance: FN = mg = 40(9.8) = 392 N up

Constant velocity: a = 0, so fk = 120 N, opposite the push

μk = fk / FN = 120 / 392 ≈ 0.31 (no units)

FN = 392 N up fk = 120 N opposite the push μk ≈ 0.31

STATIC OR SLIDING: THE 25 KG BOX

Slide 37

Given: m = 25 kg, starts at rest; μs = 0.40; μk = 0.30; pushes of 90 N and 110 N.

Find: whether the box moves for each push, and the acceleration for 110 N.

WORK / show your reasoning and units

FN = 25(9.8) = 245 N; fs,max = 0.40(245) = 98 N

90 N < 98 N: static friction is 90 N. The box stays at rest.

110 N > 98 N: the box slides. fk = 0.30(245) = 73.5 N

Fnet = 110 − 73.5 = 36.5 N; a = 36.5 / 25 = 1.46 m/s²

90 N: stays at rest 110 N: slides, a = 1.46 m/s² in the push direction

ANGLED PULL WITH FRICTION

Slide 38

Given: m = 30 kg, sliding; 150 N rope at 30° above horizontal; μk = 0.25. Positive: +x (direction of the pull).

Find: the normal force, the friction, and the acceleration.

WORK / show your reasoning and units

FN = mg – F sin 30° = 294 – 75 = 219 N

fk = 0.25(219) = 54.75 N

Fnet = 150 cos 30° – 54.75 = 75.15 N

a = 75.15 / 30 = 2.51 m/s² in +x

FN = 219 N fk = 54.75 N a = 2.51 m/s² in +x

FRICTION ON AN INCLINE

Slide 39

For a block sliding down a ramp, friction points **up the ramp**.

> **FN = mg cos θ fs,max = μsFN fk ≈ μkFN**

> **It starts sliding down if mg sin θ > μsFN.**

Friction **reverses** if sliding reverses.

SLIDING DOWN THE RAMP: THE 10 KG BLOCK

Slide 40

Given: m = 10 kg, starts at rest; 35° ramp; μs = 0.50; μk = 0.40. Positive: downhill.

Find: whether the block slides, and its acceleration.

WORK / show your reasoning and units

Fg,parallel = 10(9.8) sin 35° = 56.2 N; FN = 10(9.8) cos 35° = 80.3 N

fs,max = 0.50(80.3) = 40.1 N. 56.2 > 40.1, so it slides.

fk = 0.40(80.3) = 32.1 N

a = (56.2 – 32.1)/10 = 2.41 m/s² downhill

Slides: yes a = 2.41 m/s² downhill

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

Explain how to decide whether an object at rest will start to slide, and how to find its acceleration once it does.

Compare the push (or mg sin θ on a ramp) with fs,max = μsFN. If it is larger, the object slides: use fk = μkFN, find Fnet, then a = Fnet / m.

- [ ] I can tell static from kinetic friction and use fs,max = μsFN.
- [ ] I can find friction and acceleration on level ground and on a ramp.

Connected Objects

LEARNING TARGET I can find tension and acceleration for objects joined by cables, ropes and pulleys.

STREETLIGHT CABLE TENSION

Slide 42

Given: 20 kg streetlight at rest; left cable 30° above horizontal; right cable 60°; ignore cable mass. Positive: right and up.

Find: the tension in each support cable.

WORK / show your reasoning and units

At rest: Fnet,x = 0, Fnet,y = 0; Fdown = mg = 20(9.8) = 196 N

Horizontal: FTR cos 60° = FTL cos 30°, so FTL = 0.577FTR

Vertical: FTL sin 30° + FTR sin 60° = 196

(0.577FTR)(0.5) + 0.866FTR = 196

FTR ≈ 170 N FTL = 98.0 N

FTL = 98.0 N FTR ≈ 170 N

> **The steeper cable carries more tension.**

IDEAL STRINGS AND PULLEYS

Slide 43

Ideal: a massless, taut rope that does not **stretch**, and a massless, **frictionless** pulley.

> **Fixed-pulley systems: the rope has the same tension, T. Both objects have the same acceleration size.**

Find a using the **whole system**. Then use **one object** to find T. Draw a separate **free-body diagram** for each object.

Extra notes:

ATWOOD MACHINE

Slide 44

Given: m1 = 5 kg, m2 = 3 kg, both hanging. Positive: the motion (5 kg down, 3 kg up). **Find:** a and T.

WORK / show your reasoning and units

Fnet = (5 – 3)(9.8) = 19.6 N

Whole system: a = 19.6 / (5 + 3) = 2.45 m/s²

Use the 3 kg mass: T – 3g = 3a

T = 3(9.8 + 2.45) = 36.8 N

a = 2.45 m/s² T = 36.8 N

A BLOCK ON A FRICTIONLESS TABLE

Slide 45

Hanging mass accelerates down; table block, toward the pulley

Given: m1 = 4 kg on the table; m2 = 2 kg hanging. Positive: the motion. **Find:** a and T.

WORK / show your reasoning and units

Driving force: Fnet = m2g = 2(9.8) = 19.6 N

Whole system: a = 19.6 / (4 + 2) = 3.27 m/s²

Table block: T = m1a = 4(3.27) ≈ 13.1 N

a = 3.27 m/s² T ≈ 13.1 N

THE SAME SYSTEM WITH FRICTION

Slide 46

Given: m1 = 4 kg on the table, μk = 0.20; m2 = 2 kg hanging. Positive: the motion. **Find:** fk, a, and T.

WORK / show your reasoning and units

fk = μkm1g = 0.20(4)(9.8) = 7.84 N

Whole system: a = (19.6 – 7.84) / 6 = 1.96 m/s²

Table block: T – fk = m1a

T = 4(1.96) + 7.84 = 15.7 N

fk = 7.84 N a = 1.96 m/s² T = 15.7 N

CHALLENGE: INCLINE AND PULLEY

Slide 47

Given: m1 = 6 kg on a frictionless 30° ramp; m2 = 5 kg hanging. Positive: the motion (5 kg down, 6 kg up the ramp).

Find: a and T.

WORK / show your reasoning and units

Hanging: 5(9.8) = 49.0 N; along ramp: 6(9.8) sin 30° = 29.4 N

Whole system: a = (49.0 – 29.4) / (5 + 6) = 1.78 m/s²

Hanging mass: 5g – T = 5a

T = 5(9.8 – 1.78) ≈ 40.1 N

a = 1.78 m/s² T ≈ 40.1 N

RECALL

SECTION SUMMARY

Close your notes before you write.

Use the checklist to find what still needs practice.

Explain the two-step method for objects joined by a rope over a pulley.

Treat the whole system: a = (net driving force) / (total mass). Then write Fnet = ma for one object to find T. The tension is the same along an ideal rope.

- ☐ I can solve a cable-tension problem with Fnet,x = 0 and Fnet,y = 0.
- ☐ I can find a and T for an Atwood machine and a table-and-pulley system.

Connect the ideas, check common mix-ups, and explain the physics in your own words.

S03.1 Newton's laws

Mass (kg) never changes; weight, Fg = mg (N), depends on g. Fnet = 0 means constant velocity; otherwise Fnet = ma.

Watch out: A moving object does not need a net force to keep moving.

S03.2 Free-body diagrams

One object, one labeled arrow per force. Add with signs; split an angled force into Fx = F cos θ and Fy = F sin θ.

Watch out: Third-law pairs act on two objects, so they never cancel on one FBD.

S03.3 Forces on an incline

Tilt the axes: Fg,parallel = mg sin θ down the slope, Fg,perp = mg cos θ into the ramp, and FN = mg cos θ.

Watch out: Normal force does not always equal weight. A steeper ramp gives a smaller FN.

S03.4 Net force and acceleration

a = Fnet / m, in the direction of Fnet. Frictionless ramp: a = g sin θ. Elevator, up positive: FN – mg = ma.

Watch out: The scale reads FN, which changes with acceleration, not with moving up or down.

S03.5 Friction

Static friction matches the need, up to fs,max = μsFN. Sliding friction, fk ≈ μkFN, is usually smaller.

Watch out: Find FN first. An angled pull or a ramp changes it.

S03.6 Connected objects

Ideal rope: one tension T, one size of a. Find a from the whole system, then T from one object's Fnet = ma.

Watch out: Draw a separate FBD for each object.

QUICK RECALL / COVER THE EXPLANATIONS ABOVE

1. State the condition for equilibrium.
2. Write the weight components for a block on a ramp.
3. Explain why a scale reads more in an elevator accelerating up.